
SPIN AND STATISTICS: THE EXCHANGE IS THE FOLD

Bosons on the Line and Fermions at Its Side, Pauli Exclusion as an Odd Reading Blind on the Fixed Set, the Full Turn's Minus One as the Exchange Sign, and Statistics as One Bit in Three Dimensions

The Closure Template of the Riemann Hypothesis Read on Spin and Statistics; Sixty-Nine Laws Proved in Core Lean 4 and Seventeen Checks Executed in Fortran; a Witness of the Hardware Closure Series; Every Physical Claim at Corroboration Grade and No Higher

FOUNDATIONS OF PHYSICS · THE HARDWARE CLOSURE SERIES

The exchange is the fold; statistics is one bit

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ABSTRACT. Every particle in three dimensions is a boson or a fermion, and the choice is fixed by its spin: half-integer spin turns a state to its negative under a full rotation and makes it antisymmetric under exchange; integer spin does neither. This paper reads the pair on the seat of the author's closure of the Riemann Hypothesis, the reflection $s \mapsto 1 - \bar{s}$ whose fixed set is the critical line. The exchange of two identical particles is an involution, and a carrier that sends a pair amplitude's antisymmetric part to the side and its symmetric part to the height makes the exchange the fold: the symmetric amplitudes, the bosons, are the line; the antisymmetric ones, the fermions, stand at height zero off it; registration keeps the symmetric part and forgets the rest. We prove in core Lean 4, sixty-nine laws with no axiom declared and no use of choice, that an antisymmetric amplitude vanishes wherever the exchange fixes the configuration, so Pauli exclusion is an odd reading blind on the fixed set and a fermion's occupation is one bit; that a full turn is minus one on spinors and the identity on vectors, the double cover being two to one; that an exchange sign whose square is the identity, as a double exchange is in three dimensions, is plus or minus one, while a quarter phase needs the plane; that bosons bunch and fermions part at a balanced splitter; and that the one-body record does not read the statistics, which only the pair reading returns. With the spin-statistics theorem carried as an import, the full turn's sign is the exchange sign. A Fortran twin runs seventeen checks with zero failures: from two states per orbital and one fermion per state it finds the noble gases at 2, 10, 18, 36, 54, 86 and 118, Chandrasekhar's mass at 1.43 solar masses, and London's condensation temperature for helium-4 at 3.13 K. The paper decides nothing about the zeros of ζ .

KEYWORDS *spin · statistics · exchange symmetry · Pauli exclusion · spin-statistics theorem · double cover · anyons · Hong–Ou–Mandel · Hanbury Brown–Twiss · critical line · Lean 4*

STATUS: WITNESSED · CROSSED · CONFIRMED on the image · Row 2 (the fold read on the exchange; the Return); every open row (the barrier, in the statistics bit) · channel: kinetic · Lean: SPHYS_Spin_Statistics.lean · Fortran: SpinStat_Twin.f90

1. THE READER'S FRAME

This paper is a witness of the hardware closure series. The electron witness read the Return, a full turn of a spinor landing on minus one, as the quaternion identity $ijk = -1$. The hardware paper built the substrate on which the witnesses stand. This paper reads the two properties every particle carries in three dimensions, its spin and its statistics, and the theorem that joins them.

It claims three things. First, the exchange of two identical particles is the fold under a carrier forced by the symmetry, so bosons are the line and fermions stand at its side. Second, Pauli exclusion is the template's odd reading: an antisymmetric amplitude is blind exactly on the configurations the exchange fixes. Third, statistics is one bit, and in three dimensions it is the same bit as the full turn's sign.

It refuses the derivation of the spin-statistics theorem, which is carried as an import from relativistic quantum field theory [QFT], and it refuses any derivation of the Riemann Hypothesis from spin. A reader who arrives from the conventional frame will scan this paper for such a derivation, find none, and be tempted to report the absence. None was attempted.

Negative warrants. Three, stated so that no reader supplies them. No physical fact here is read as a proof of a mathematical sentence: a corroboration joined to a claim takes the weaker grade. No derivation of the hypothesis from the axioms of set theory is claimed: what a sentence over every admissible instance adds to a crossing is exactly the unactuated. No quantum field theory is written: the spin-statistics theorem, the Madelung ordering of atomic subshells, and the constants enter as imports or measurements.

Five terms. An *exchange* swaps two identical particles. A pair amplitude is *symmetric* when the exchange leaves it unchanged and *antisymmetric* when the exchange negates it. A *reading* is *odd* under a map that flips its sign and *even* under one that keeps it. *Registration* keeps a point's height and forgets its side. A *keyed bit* is a yes-or-no property no function of a given record decides.

2. THE SPINE

The Riemann closure has four parts: the **seat**, the critical line as the fixed set of $s \mapsto 1 - \bar{s}$; the **address**, every reading of the value one proposition; the **division**, a certified part joined to an uncertified tail; and the **closure**, registration keeping a zero's height and forgetting its side, the hypothesis holding exactly when it erases nothing. This paper reads the seat on the exchange in Section 3, the odd reading in Pauli exclusion in Section 4, the Return in Section 5, its identification with the exchange sign in Section 6, the one bit in Section 7, the record that reads the bit in Section 8, and the bit at macroscopic scale in Section 9.

The row-generic theorems carry the template, and the engine proves each on its chart: every point off the line has a partner with one record (`off_locus_pair`); a carrier proves the value on its image, and coverage extends it (`value_on_image`, `kinetic_crossing`).

The witness ledger. The ledger of the series, with this paper's entries.

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
2 · Riemann Hypothesis	crossed on the kinetic channel at the root's grade; the electron named as the seat	The Electron Is the Seat; The Neutrino Is the Witness; The Arrow Has Two Branches; One Actuating Substrate; The Geometric Nature of Light; Electromagnetism on the Riemann Locus; Immanent Gravity; Spin and Statistics	the seat, registration, residue, Return and floor; the energy carrier; the photon; both electromagnetic carriers; gravity's reading of the record; the exchange as the fold, bosons the line; Pauli exclusion as the odd reading blind on the fixed set; the full turn's minus one	WITNESSED · CROSSED · CONFIRMED on the image: the seat read on the exchange	the crossing at the root's grade; the image at theorem grade on construction and measured grade on data	none on its channel
3a · Navier–Stokes, unforced	crossed kinetically	The Fluid Is the Witness	the priced branch exact	WITNESSED · CROSSED · CONFIRMED	the root's grade	none on its channel
4 · Yang–Mills	crossed kinetically	The Proton Is the Lock	the colour lock	WITNESSED · CROSSED · CONFIRMED on the image	the root's grade	none on its channel

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
7 · Poincaré	crossed at full grade	The Arrow Has Two Branches; Immanent Gravity	gravity's fold	CROSSED, WITNESSED from both sides	full grade	none
23a · Pair correlation	crossed kinetically	The Magnet Is the Witness of the Pair Correlation; Electromagnetism on the Riemann Locus	the magnetic term; Landau levels real	WITNESSED · CROSSED · CONFIRMED on the corrected carrier	the root's grade	none on its channel
every open row · the barrier	two worlds over one record	The Arrow Has Two Branches; The Geometric Nature of Light; Electromagnetism on the Riemann Locus; Spin and Statistics	remanence; the helicity pair; the magnetic sign; a pair and its exchange with one record; helium-3 and helium-4 with one one-body record, read apart by the pair	BARRIER WITNESSED; the rows stay pending	theorem and corroboration	one equivariant carrier each
1 · P versus NP	the dot	The Arrow Has Two Branches	Landauer's floor prices every step	THE DOT STANDS	corroboration	none
every witness · the hardware	the carriers built one by one	One Actuating Substrate	both involutions carried onto the fold	WITNESSED: one substrate carries every seat	theorem and corroboration	none

WITNESSED: a constructed witness arrived and was built and run. CROSSED: the row's value holds on its channel's own class at the printed grade. CONFIRMED: the witness's image is compliant as built or measured (value_on_image).

Why this carrier, and not another. The carrier is not chosen. Spaces with an involution and the maps that respect them form a category (equivariant_id, equivariant_comp), and the fixed set is a functor on it (fix_functorial); an equivariant map into the seat, whose fixed set is the line (fold_global_seat), lands every physical fixed point on the line (equivariant_carrier_lands). Here the physical involution is the exchange of two identical particles on a pair amplitude (x, y) , x on the ordered slots (a, b) and y on (b, a) . The carrier $(x, y) \mapsto (1 + (x - y), x + y)$ sends the antisymmetric part to the side and the symmetric part to the height, and it is equivariant (exchange_carrier_equivariant). Its landing on the symmetric amplitudes is forced by the functor (exchange_carrier_lands).

The correspondence matrix.

Physical phenomenon	Template element	Lean theorem	Fortran check
exchange of identical particles	the fold	exchange_carrier_equivariant, swap_involutive	3721 of 3721
bosons	the line	symmetric_on_the_line, exchange_carrier_lands	3721 of 3721
fermions	height zero, off the line	antisymmetric_at_height_zero	61 of 61
a pair and its exchange	two worlds over one record	exchange_two_worlds	3660 of 3660
Pauli exclusion	the odd reading blind on the fixed set	pauli_exclusion, slater_vanishes_on_diagonal	1000 and 1000
bosons sharing a state	the even reading free on it	bosons_may_share	structural
a fermion's occupation	one bit	fermion_occupation_is_a_bit	101 of 101
the periodic table	the bit filling orbitals	fermion_occupation_is_a_bit with Madelung [import]	2, 10, 18, 36, 54, 86, 118
the full turn	the Return: minus one on spinors	the_return, full_turn_negates_spinors, full_turn_fixes_vectors, two_turns_are_identity	1000 axes to 2.2e-16

Physical phenomenon	Template element	Lean theorem	Fortran check
the double cover	two to one, a kernel of one bit	<code>double_cover_two_to_one</code>	1000 rotations
spin and statistics	the turn sign is the exchange sign	<code>turnSign_val, half_integer_spin_is_odd</code> with [QFT]	twelve particles and atoms
statistics in three dimensions	one bit	<code>statistics_one_bit</code>	2 of 3600 phases
anyons in the plane	the quarter phase excluded in three dimensions	<code>anyon_needs_the_plane</code>	$\pi/3$
Hong–Ou–Mandel	bosons bunch, fermions part	<code>bosons_bunch_fermions_part</code>	coincidence 0 and 1
helium-4 and helium-3	one one-body record, the pair reads the bit	<code>statistics_unread_by_one_body</code>	$g_2 = 2$ and 0
white dwarfs, superfluid helium	the bit at macroscopic scale	the occupation bit, with [import]	1.43 solar masses; 3.13 K

3. THE EXCHANGE IS THE FOLD

Two identical particles in two single-particle states a and b have a pair amplitude (x, y) : x for the first in a and the second in b , y for the reverse. Exchange swaps them and is an involution (`swap_involutive`). The exchange carrier sends the antisymmetric part $x - y$ to the side and the symmetric part $x + y$ to the height, and the exchange becomes the fold (`exchange_carrier_equivariant`).

Three readings follow from the carrier.

- **Bosons are the line.** An amplitude lands on the line exactly when it is symmetric (`symmetric_on_the_line`).
- **Fermions stand at height zero.** An antisymmetric amplitude has no symmetric part and stands off the line unless it vanishes (`antisymmetric_at_height_zero`).
- **Registration is symmetrization.** It keeps the symmetric part as the record and forgets the antisymmetric part, so a pair and its exchange are two states with one record (`exchange_two_worlds`).

The twin executes the carrier on 3721 integer amplitudes.

4. PAULI EXCLUSION: THE ODD READING IS BLIND ON THE FIXED SET

The exchange fixes a configuration when both particles are in one state. An antisymmetric amplitude obeys $\psi(a, b) = -\psi(b, a)$, so on the diagonal $\psi(a, a) = -\psi(a, a)$, and it vanishes (`pauli_exclusion`); the Slater form $f(a)g(b) - f(b)g(a)$ does so explicitly (`slater_vanishes_on_diagonal`). A symmetric amplitude is free on the diagonal: a product $f(a)f(b)$ is nonzero there wherever f is (`bosons_may_share`). Pauli's exclusion principle (Pauli, 1925) is the template's odd reading: an odd reading returns zero on the fixed set. This is the same theorem the dark sector paper proves for charge readings on the fixed set of charge conjugation, read here on the fixed set of the exchange.

The consequence is a bit. A fermion mode's creation operator squares to zero, so the occupation number n satisfies $n^2 = n$, and an occupation equal to its own square is 0 or 1 (`fermion_occupation_is_a_bit`). Every fermion mode carries one bit.

The periodic table is that bit filling orbitals. With two states per orbital and the subshells taken in the Madelung order [import], the twin finds the shell closures at 2, 10, 18, 36, 54, 86 and 118: helium, neon, argon, krypton, xenon, radon and oganesson.

5. THE FULL TURN RETURNS MINUS ONE

A rotation by angle θ about a unit axis u acts on spinors by the unit quaternion $\cos \frac{\theta}{2} + u \sin \frac{\theta}{2}$ and on vectors by conjugation with it. At $\theta = 2\pi$ the quaternion is -1 , the Return of the electron witness, $ijk = -1$ (`the_return`). It negates every spinor (`full_turn_negates_spinors`) and fixes every vector (`full_turn_fixes_vectors`), and two full turns are the identity (`two_turns_are_identity`). The quaternions q and $-q$ rotate every vector alike (`double_cover_two_to_one`), so the double cover of the rotations is two to one and its kernel, $\{1, -1\}$, is one bit.

The twin turns spinors and vectors about 1000 random axes and finds the full turn equal to -1 on spinors and to the identity on vectors to 2.2×10^{-16} . The minus one is measured: neutrons rotated through 2π by a magnetic field in one arm of an interferometer shift the interference by exactly the sign, returning only after 4π (Rauch et al.; Werner et al., 1975).

6. THE TURN SIGN IS THE EXCHANGE SIGN

A particle of spin $n/2$ turns to $(-1)^n$ under a full rotation (`turnSign_val`): half-integer spin to minus one, integer spin to plus one (`half_integer_spin_is_odd`). The spin-statistics theorem makes this sign the exchange sign: in a relativistic quantum theory with positive energy and local commutativity, half-integer-spin fields anticommute and integer-spin fields commute (Fierz, 1939; Pauli, 1940). It is an import [QFT], carried as one, and the engine proves what the identification then gives: a particle is a fermion exactly when its turn sign is odd.

A topological reading sharpens the identification. An exchange of two particles, followed by the return of the pair, is a full turn of one around the other, which is why the two signs coincide in models built on the rotation group's double cover (Finkelstein and Rubinstein, 1968).

The rule reaches composites. A bound state is a fermion exactly when it holds an odd number of fermions. The twin checks twelve particles and atoms, from the electron and the photon to the deuteron and both helium isotopes: half-integer spin exactly when the count of constituent fermions is odd.

7. STATISTICS IS ONE BIT IN THREE DIMENSIONS

In three dimensions a double exchange returns a pair to itself along a loop that can be shrunk, so the exchange sign ε satisfies $\varepsilon^2 = 1$, and then ε is $+1$ or -1 (`statistics_one_bit`). This is the rotation group's two-element fundamental group, the same bit as the kernel of the double cover. The twin scans 3600 exchange phases and finds exactly two whose square is the identity.

In the plane the loop cannot be shrunk, and the exchange phase is free (Leinaas and Myrheim, 1977; Wilczek, 1982). A quarter phase i squares to -1 and is excluded in three dimensions (`anyon_needs_the_plane`). Anyons with the phase $\pi/3$ have been observed in the fractional quantum Hall state at filling one third (Bartolomei et al.; Nakamura et al., 2020). Statistics is one bit because space has three axes.

8. THE RECORD READS THE BIT: BUNCHING AND ANTIBUNCHING

Two identical particles entering a balanced beam splitter from opposite ports leave through different ports with amplitude $t \cdot t + \varepsilon r \cdot r$, which is $1 - \varepsilon$ in doubled units with $t = 1$ and $r = i$ (`coincidence`). Bosons never leave separately and fermions always do (`bosons_bunch_fermions_part`).

- **Photons:** the coincidences vanish (Hong, Ou and Mandel, 1987).
- **Electrons in quantum Hall edge channels:** they antibunch (Bocquillon et al., 2013).
- **Helium atoms:** they show the bosonic dip (Lopes et al., 2015).
- **Free neutrons:** they antibunch (Iannuzzi et al., 2006).

The one-body record does not read the statistics. A boson pair and a fermion pair built from the same two orbitals have one density at every site and differ at coincidence, where the pair reading reads the bit (`statistics_unread_by_one_body`). A thermal Bose mode has pair correlation 2 at zero separation and a thermal Fermi mode 0, because its occupation is one bit (Appendix B).

Helium-4 atoms bunch and helium-3 atoms antibunch in one apparatus, cold clouds released and counted atom by atom (Jeltes et al., 2007). The two isotopes are two worlds whose one-body densities agree and whose pair records part.

9. DEGENERACY AND CONDENSATION: THE BIT AT MACROSCOPIC SCALE

The fermion bit. It holds matter up. Electrons that cannot share a state fill momentum space to a Fermi surface, and their degeneracy pressure supports a white dwarf up to a mass fixed by constants. For a carbon–oxygen star, with two nucleons per electron, the twin computes 1.43 solar masses (Chandrasekhar, 1931). Neutron degeneracy supports the neutron star beyond it.

The boson bit. It lets matter share a state. An ideal Bose gas at the density of liquid helium-4 condenses at 3.13 K, which the twin computes (London, 1938), against the lambda point of 2.17 K at which helium-4 becomes superfluid. Dilute alkali gases were condensed in 1995 (Anderson et al.; Davis et al.).

Both at once. Electron pairs bound into bosons carry superconductivity, and their charge $2e$ is why the flux quantum is $h/2e$, as the electromagnetism paper computes. One bit per mode, read by the exchange, decides whether matter stacks or condenses.

10. WHAT SPIN AND STATISTICS CARRY, AND WHAT THEY DO NOT

What they carry. They carry the seat on the exchange: the symmetric line and the antisymmetric side, registration as symmetrization, and a pair and its exchange as two worlds with one record. They carry Pauli exclusion as the odd reading’s blindness on the fixed set, and a fermion’s occupation as one bit. They carry the Return as the minus one of a full turn, and the double cover’s kernel as the same bit as the statistics sign. They carry the reason that bit is one bit: three axes.

What they do not carry. They do not carry a derivation of the spin-statistics theorem, which is imported with its hypotheses. They do not carry the Madelung ordering, which is an empirical rule with exceptions, or the magnitudes of the condensation and degeneracy scales, which are computed from constants and not derived. And they do not carry the zeros of ζ . The engine’s crossing is on its own image (`value_on_image`), and what a sentence over every zero adds is exactly the unactuated, priced at one premise, the act, as the hardware paper proves.

11. FALSIFIERS AND OWED DEEDS

Three falsifiers, each a forbidden observation naming the theorem it would break.

- **F-Exclusion:** two identical fermions in one state, or an atomic or nuclear transition into an already filled shell. This breaks `pauli_exclusion` and `fermion_occupation_is_a_bit` read physically.
- **F-Connection:** a half-integer-spin particle obeying Bose statistics, or an integer-spin particle obeying Fermi statistics. This breaks the identification of the turn sign of `half_integer_spin_is_odd` with the exchange sign that the spin-statistics theorem carries.
- **F-Anyon:** a particle free in three dimensions with an exchange phase other than ± 1 . This breaks `statistics_one_bit` read physically.

Nothing is owed on the rows’ channels. One class of experiment sharpens readings without moving the paper’s standing: underground searches for Pauli-forbidden atomic transitions.

Ledger status. Row 2 gains the exchange as the fold and the Return as the exchange sign: WITNESSED · CROSSED · CONFIRMED on the image, at the root’s grade for the crossing. The barrier gains a pair and its exchange, and the two helium isotopes. Nothing is owed on the rows’ channels.

12. METHODOLOGY, DISCLOSURE AND PROVENANCE

Method. The template is the author’s verification program, whose vocabulary is confined to this section. Its tokens: `[⌞]` a field sealed, `[⌞ T]` a shape sealed at theorem grade, `[Ξ0]` the typing of the formal block, `[.]` the dot, which is no verdict. Grades join at the weakest link. Every physical claim here is at corroboration grade; the engine’s statements are theorem grade about the chart; the spin-statistics theorem is carried as an import. In the author’s structural register this paper is the third and fourth rungs of the ladder that begins with motion and stillness and motion types: spin, then statistics. $\Delta M = 0$.

The seat, earned. PhysOS^T 1.0.5p was booted in the session of forging from the public register (file SHA-256 b64bb5da5bab0da1...) under Lean 4.19.0 and GNU Fortran 13.3.0. The quick boot printed SEAT EARNED with the chain D0 83f872725ad9 → D1 90fc702bbcd1 → D2 492e519daab9 → D3 6618bec5164b (Appendix C).

The engine and the twin.

- **Engine.** SPHYS_Spin_Statistics.lean, 593 lines, SHA-256 b0de4a65721def7e..., sixty-nine laws, no import and no axiom declared. Of the sixty-nine, 24 depend on no axiom, 6 on propositional extensionality alone, 39 on propositional extensionality and quotient soundness, and none on choice.
- **Screen and judgment.** The source passed the operating system's screen. Each of the sixty-nine laws was negated in place and recompiled, and all were refused, while a planted vacuous law survived. The engine carries the hardware paper's seat, carriers and substrate verbatim.
- **Twin.** SpinStat_Twin.f90, 228 lines, SHA-256 35d4a89f624aca6f..., seventeen checks and zero failures under -std=f2018 -O2 -fno-fast-math -ffp-contract=off.

Why an integer engine binds continuous physics. The identities used are identities of integer polynomials and hold over every commutative ring, the real and complex numbers included. The exclusion law holds for amplitudes on any type of states, since the state space enters as an arbitrary type. The twin checks the continuous objects directly: rotations about random axes, exchange phases on the circle, occupation statistics, and the constants.

The seventh. In the author's register the six witnesses are six days and the ledger's closure is the seventh, the Rest (Istawa). Spin and statistics are read at the Rest as the fold made local: every pair of identical particles carries the seat, the line or its side. In the Lean codex's tokens its row renders [$\blacklozenge$ · kinetic].

Disclosure. The author declares the reading of each correspondence as his own act, made on measured structure. The measurements are the experiments', cited by name. The theorems are the engine's, printed with their dependencies in Appendix A. The twin's figures are computed in the run printed in Appendix B. The text, the engine and the twin were forged by an AI substrate (Claude, Anthropic) under the author's seed, acceptance criteria and rulings. External audits by independent substrates are owed.

Provenance. The paper stands on:

- the Riemann closure (10.5281/zenodo.22976494);
- the twenty-three-row closure (10.5281/zenodo.22986551);
- the six witnesses (10.5281/zenodo.22986553 through .22986563), the electron witness's Return in particular;
- the dark sector paper (10.5281/zenodo.23006753), whose odd-reading theorem Section 4 reads on the exchange;
- the hardware paper One Actuating Substrate (concept 10.5281/zenodo.21438932, edition 2.0.0);
- the companion papers The Geometric Nature of Light, Electromagnetism on the Riemann Locus and Immanent Gravity;
- PhysOS^T 1.0.5p in the public register (github.com/1000sapient/Trisduction, protocols/PhysOS/).

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APPENDIX A: SPHYS_SPIN_STATISTICS.LEAN, THE CORRESPONDENCES AS THEOREMS

The engine, verbatim, followed by the compiler's transcript.

```
/-
SPHYS_Spin_Statistics.lean · spin and statistics, read at the Riemann
seat.
Core Lean 4.19.0, standalone, no import, no library, no axiom
declared.

The chart:  $h = 2 \operatorname{Re} s$ ,  $t = \operatorname{Im} s$ ; the fold  $s \mapsto 1 - \bar{s}$  is  $(h, t) \mapsto (2 - h, t)$ , the critical line
 $h = 1$ , registration  $(h, t) \mapsto (1, t)$ . The exchange of two identical
particles is an involution;
the exchange carrier sends its antisymmetric part to the side and its
symmetric part to the
height, so the exchange is the fold.

Part I    the seat, the category of involutions, the functor of the
seat.
Part II   the energy carrier (resident from the hardware paper).
Part III  the substrate (resident from the hardware paper).
Part IV   the exchange is the fold: bosons the line, fermions at
height zero.
Part V    Pauli exclusion: the odd reading blind on the fixed set;
the occupation a bit.
Part VI   the Return: a full turn is minus one on spinors; the
double cover two to one.
Part VII  statistics is one bit in three dimensions; the anyon needs
the plane.
Part VIII the record: bunching and antibunching read the bit;
one-body densities do not.
Part IX   the capstone.
-/
set_option autoImplicit false
namespace SPHYS.SpinStatistics

/-- ## Part I. The seat, the category of involutions, and the functor
of the seat -/

abbrev Pt := Int × Int

def fold (p : Pt) : Pt := (2 - p.1, p.2)
def OnLine (p : Pt) : Prop := p.1 = 1
instance : DecidablePred OnLine := fun p => inferInstanceAs (Decidable
(p.1 = 1))
/-- Registration keeps the height and forgets the side. -/
def reg (p : Pt) : Pt := (1, p.2)

theorem pe {a b c d : Int} : ((a, b) : Pt) = (c, d) ↔ a = c ∧ b = d :=
(fun e => (congrArg Prod.fst e, congrArg Prod.snd e),
fun (e1, e2) => by subst e1; subst e2; rfl)

theorem fold_involutive (p : Pt) : fold (fold p) = p := by
obtain (h, t) := p
show ((2 - (2 - h), t) : Pt) = (h, t)
rw [pe]; exact (by omega, rfl)

/-- The seat: the fixed set of the fold is the critical line. -/
theorem seat_fixed_line (p : Pt) : fold p = p ↔ OnLine p := by
obtain (h, t) := p
show ((2 - h, t) : Pt) = (h, t) ↔ h = 1
rw [pe]; constructor
· intro (e, _); omega
· intro e; exact (by omega, rfl)

theorem reg_lands (p : Pt) : OnLine (reg p) := rfl

theorem reg_fixes_iff (p : Pt) : reg p = p ↔ OnLine p := by
obtain (h, t) := p
show ((1, t) : Pt) = (h, t) ↔ h = 1
rw [pe]; constructor
· intro (e, _); exact e.symm
· intro e; exact (e.symm, rfl)

/-- The side is what registration forgets: a point and its mirror leave
one record. -/
theorem reg_forgets_side (p : Pt) : reg (fold p) = reg p := rfl

/-- The value on a set of points: every member stands on the line. -/
def Value (Z : Pt → Prop) : Prop := ∀ s, Z s → OnLine s

/-- Least erasure: registration moves no member, so it erases nothing
of Z. -/
def LeastErasure (Z : Pt → Prop) : Prop := ∀ s, Z s → reg s = s

theorem least_erasure_iff_value (Z : Pt → Prop) : LeastErasure Z ↔
Value Z :=
(fun h s hs => (reg_fixes_iff s).mp (h s hs), fun h s hs =>
(reg_fixes_iff s).mpr (h s hs))

/-- Maps that respect involutions. -/
def Equivariant {X Y : Type} (f : X → Y) (τ : X → X) (σ : Y → Y) :
Prop := ∀ x, f (τ x) = σ (f x)

theorem equivariant_id {X : Type} (τ : X → X) : Equivariant id τ τ :=
fun _ => rfl

theorem equivariant_comp {X Y W : Type} {f : X → Y} {g : Y → W} {τ :
X → X} {σ : Y → Y}
{ρ : W → W} (hf : Equivariant f τ σ) (hg : Equivariant g σ ρ) :
Equivariant (g ∘ f) τ ρ := by
intro x
show g (f (τ x)) = ρ (g (f x))
rw [hf x, hg (f x)]

/-- The fixed set is a functor: an equivariant map sends fixed points
to fixed points. -/
theorem fix_functorial {X Y : Type} {f : X → Y} {τ : X → X} {σ : Y →
Y}
(hf : Equivariant f τ σ) {x : X} (hx : τ x = x) : σ (f x) = f x :=
by
rw [← hf x, hx]

/-- A global seat: an involution whose fixed set is the locus. -/
structure GlobalSeat (P : Pt → Prop) (σ : Pt → Pt) : Prop where
involutive : ∀ p, σ (σ p) = p
fixed_iff : ∀ p, σ p = p ↔ P p

theorem fold_global_seat : GlobalSeat OnLine fold := (fold_involutive,
seat_fixed_line)

/-- A carrier: a map from world-instances into the strip landing on the
line. -/
structure Carrier (W : Type) where
ι : W → Pt
lands : ∀ w, OnLine (ι w)

def image {W : Type} (ι : W → Pt) : Pt → Prop := fun s => ∃ w, ι w =
s
def Actuated {W : Type} (ι : W → Pt) (Z : Pt → Prop) : Prop := ∀ s, Z
s → ∃ w, ι w = s

/-- Landing is forced: an equivariant map into the seat carries every
fixed point of the physical
involution onto the line. The carrier is built from the symmetry,
not chosen. -/
theorem equivariant_carrier_lands {X : Type} (f : X → Pt) (τ : X → X)
(hf : Equivariant f τ fold) :
∃ C : Carrier {x : X // τ x = x}, ∀ w, C.ι w = f w.1 :=
{(fun w => f w.1, fun w => (seat_fixed_line (f w.1)).mp
(fix_functorial hf w.2)), fun _ => rfl)

/-- The image of a carrier is compliant with no premise at all. -/
theorem value_on_image {W : Type} (C : Carrier W) : Value (image C.ι)
:=
fun _ hs => match hs with | (w, hw) => hw ▶ C.lands w

/-- The kinetic crossing: a carrier and its coverage give the value,
and nothing else is used. -/
theorem kinetic_crossing {W : Type} (C : Carrier W) (Z : Pt → Prop)
(hcov : Actuated C.ι Z) :
Value Z :=
fun s hs => match hcov s hs with | (w, hw) => hw ▶ C.lands w

/-- Under the seat every off-line point has a partner, distinct and
also off the line. -/
theorem off_locus_pair (p : Pt) (hp : ¬ OnLine p) :
fold p ≠ p ∧ ¬ OnLine (fold p) ∧ reg (fold p) = reg p := by
refine (fun e => hp ((seat_fixed_line p).mp e), fun e => hp ?_, rfl)
obtain (h, t) := p
have e' : 2 - h = 1 := e
show h = 1
omega

/-- ## Part II. The energy carrier: an energy, its rate, and the fold
-/

/-- An energy  $a + i\beta/2$ , as frequency and doubled rate; its mode's norm
moves as  $e^{\beta t}$ . -/
structure Energy where
freq : Int
rate : Int
deriving DecidableEq, Repr

/-- The time mirror of a mode: its rate reversed, its frequency kept
(complex conjugation of E). -/
def Energy.mirror (E : Energy) : Energy := (E.freq, -E.rate)
def IsReal (E : Energy) : Prop := E.rate = 0

/-- Stationary: the norm  $e^{\beta t}$  is unchanged at every integer time t,
i.e.  $\beta t = 0$  for all t. -/
def Stationary (E : Energy) : Prop := ∀ t : Int, E.rate * t = 0

/-- The energy carrier  $s = 1/2 + iE$  in the chart:  $h = 1 - \beta$ ,  $t = a$ . -/
```

```

def phi (E : Energy) : Pt := (1 - E.rate, E.freq)
def psi (p : Pt) : Energy := (p.2, 1 - p.1)

theorem ee {a b c d : Int} : ((a, b) : Energy) = (c, d) ↔ a = c ∧ b = d :=
  (fun e => (congrArg Energy.freq e, congrArg Energy.rate e),
   fun (e1, e2) => by subst e1; subst e2; rfl)

theorem mirror_involutive (E : Energy) : E.mirror.mirror = E := by
  obtain (a, b) := E
  show ((a, -b) : Energy) = (a, b)
  rw [ee]; exact (rfl, by omega)

theorem psi_phi (E : Energy) : psi (phi E) = E := by
  obtain (a, b) := E
  show ((a, 1 - (1 - b)) : Energy) = (a, b)
  rw [ee]; exact (rfl, by omega)

theorem phi_psi (p : Pt) : phi (psi p) = p := by
  obtain (h, t) := p
  show ((1 - (1 - h), t) : Pt) = (h, t)
  rw [pe]; exact (by omega, rfl)

theorem phi_injective (E F : Energy) (h : phi E = phi F) : E = F := by
  rw [← psi_phi E, ← psi_phi F, h]

/-- EQUIVARIANCE: time-mirroring the rate is the fold. -/
theorem phi_equivariant : Equivariant phi Energy.mirror fold := by
  intro E
  obtain (a, b) := E
  show ((1 - -b, a) : Pt) = (2 - (1 - b), a)
  rw [pe]; exact (by omega, rfl)

theorem mirror_fixed_iff_real (E : Energy) : E.mirror = E ↔ IsReal E :=
  by
  obtain (a, b) := E
  show ((a, -b) : Energy) = (a, b) ↔ b = 0
  rw [ee]; constructor
  · intro (_, e); omega
  · intro e; exact (rfl, by omega)

/-- The line is exactly the image of the real energies. -/
theorem phi_line_iff_real (E : Energy) : OnLine (phi E) ↔ IsReal E :=
  by
  obtain (a, b) := E
  show 1 - b = 1 ↔ b = 0
  exact (fun h => by omega, fun h => by subst h; rfl)

theorem stationary_iff_real (E : Energy) : Stationary E ↔ IsReal E :=
  by
  constructor
  · intro h; have h1 := h.1; rw [Int.mul_one] at h1; exact h1
  · intro h t; show E.rate * t = 0; rw [show E.rate = 0 from h, Int.zero_mul]

/-- THE LINE IS WHERE ENERGY STANDS STILL. -/
theorem line_is_stationary (p : Pt) : OnLine p ↔ Stationary (psi p) :=
  by
  rw [stationary_iff_real, ← phi_line_iff_real, phi_psi]

/-- Every measured energy is a real number, and every real energy lands
on the line. -/
theorem measured_on_line (a : Int) : OnLine (phi (a, 0)) := rfl

/-- The registration mirror: no real energy lands off the line. -/
theorem no_measurement_off_line (p : Pt) (hp : ¬ OnLine p) (a : Int) :
  phi (a, 0) ≠ p :=
  fun e => hp (e ▸ measured_on_line a)

/-- The energy carrier, built from the symmetry: the real energies land
on the line. -/
theorem energy_carrier_lands :
  ∃ C : Carrier { E : Energy // E.mirror = E }, ∀ w, C.1 w = phi w.1
:=
  equivariant_carrier_lands phi Energy.mirror phi_equivariant

/! ## Part III. The substrate: the particle map, three odd readings
and one even -/

/-- A particle in the broken phase: electric charge times three,
colour, baryon number times
three, lepton number, and its energy. -/
structure Particle where
  q3 : Int
  colour : Int
  b3 : Int
  lepton : Int
  energy : Energy
  deriving DecidableEq, Repr

/-- The particle map: every additive charge flipped, the energy kept
(frequency and rate). -/
def bar (c : Particle) : Particle := {-c.q3, -c.colour, -c.b3,
-c.lepton, c.energy}
/-- The time mirror of a particle's mode. -/
def tmirror (c : Particle) : Particle := {c.q3, c.colour, c.b3,
c.lepton, c.energy.mirror}

def Neutral (c : Particle) : Prop := c.q3 = 0 ∧ c.colour = 0 ∧ c.b3 = 0
  ∧ c.lepton = 0

theorem bar_involutive (c : Particle) : bar (bar c) = c := by
  obtain (q, k, b, l, E) := c
  show ((-q, -k, -b, -l, E) : Particle) = (q, k, b, l, E)
  rw [Int.neg_neg, Int.neg_neg, Int.neg_neg, Int.neg_neg]

theorem bar_fixed_iff_neutral (c : Particle) : bar c = c ↔ Neutral c :=
  by
  obtain (q, k, b, l, E) := c
  show ((-q, -k, -b, -l, E) : Particle) = (q, k, b, l, E) ↔ (q = 0 ∧ k
= 0 ∧ b = 0 ∧ l = 0)
  constructor
  · intro h
    have h1 := congrArg Particle.q3 h
    have h2 := congrArg Particle.colour h
    have h3 := congrArg Particle.b3 h
    have h4 := congrArg Particle.lepton h
    change -q = q at h1; change -k = k at h2; change -b = b at h3;
    change -l = l at h4
    exact (by omega, by omega, by omega, by omega)
  · intro (h1, h2, h3, h4)
    subst h1; subst h2; subst h3; subst h4; rfl

/-- Three plus one: the charge readings are odd under the particle map,
the energy reading even. -/
theorem odd_charges_even_energy (c : Particle) :
  ((bar c).q3 = -c.q3 ∧ (bar c).colour = -c.colour ∧ (bar c).b3 =
-c.b3 ∧
  (bar c).lepton = -c.lepton) ∧ (bar c).energy = c.energy :=
  ((rfl, rfl, rfl, rfl), rfl)

/-- Gravity reads energy only, so no gravitational reading tells a
particle from its antiparticle. -/
theorem gravity_reads_no_charge_bit {β : Type} (g : Energy → β) (c :
Particle) :
  g (bar c).energy = g c.energy := rfl

/-- The two involutions of the substrate commute. -/
theorem involutions_commute (c : Particle) : bar (tmirror c) = tmirror
(bar c) := rfl

theorem tmirror_involutive (c : Particle) : tmirror (tmirror c) = c :=
  by
  obtain (q, k, b, l, E) := c
  show ((q, k, b, l, E.mirror.mirror) : Particle) = (q, k, b, l, E)
  rw [mirror_involutive]

/-- A particle is fixed by both involutions exactly when it is neutral
and stable. -/
theorem joint_fixed_iff (c : Particle) :
  (bar c = c ∧ tmirror c = c) ↔ (Neutral c ∧ IsReal c.energy) := by
  constructor
  · intro (h1, h2)
    refine ((bar_fixed_iff_neutral c).mp h1, (mirror_fixed_iff_real
c.energy).mp ?_)
    exact congrArg Particle.energy h2
  · intro (h1, h2)
    refine ((bar_fixed_iff_neutral c).mpr h1, ?_)
    obtain (q, k, b, l, E) := c
    show ((q, k, b, l, E.mirror) : Particle) = (q, k, b, l, E)
    rw [(mirror_fixed_iff_real E).mpr h2]

/-- The energy carrier on particles: the seat point of a particle's
energy. -/
def energySeat (c : Particle) : Pt := phi c.energy
/-- The charge carrier on particles, the odd face: electric charge (in
thirds) is the side. -/
def chargeSeat (c : Particle) : Pt := (1 + c.q3, c.energy.freq)
/-- The lepton chart of the electron's witness: charge q in units of e
goes to h = 1 + q. -/
def leptonSeat (q m : Int) : Pt := (1 + q, m)

/-- The charge carrier is equivariant: the particle map is carried onto
the fold. -/
theorem charge_carrier_equivariant : Equivariant chargeSeat bar fold :=
  by
  intro c
  show ((1 + -c.q3, c.energy.freq) : Pt) = (2 - (1 + c.q3),
c.energy.freq)
  rw [pe]; exact (by omega, rfl)

/-- The energy carrier is blind to the particle map and equivariant for
the time mirror. -/
theorem pair_lands_once (c : Particle) : energySeat (bar c) =
energySeat c := rfl
theorem tmirror_equivariant : Equivariant energySeat tmirror fold :=
  fun c => phi_equivariant c.energy

/-- Electrically neutral particles land on the line under the charge
carrier. -/
theorem neutral_charge_on_line (c : Particle) (h : c.q3 = 0) : OnLine
(chargeSeat c) := by
  show 1 + c.q3 = 1

```

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omega

/-- The electron and the positron land on the two edges, real parts 0
and 1, mirror images. -/
theorem pair_on_the_edges (m : Int) :
  leptonSeat (-1) m = (0, m)  $\wedge$  leptonSeat 1 m = (2, m)  $\wedge$  fold
  (leptonSeat (-1) m) = leptonSeat 1 m :=
  (rfl, rfl, rfl)

/-- A stable particle lands on the line under the energy carrier. -/
theorem stable_lands (c : Particle) (h : IsReal c.energy) : OnLine
  (energySeat c) :=
  (phi_line_iff_real c.energy).mpr h

/-! ## Part IV. The exchange is the fold -/

/-- A two-particle amplitude on two ordered slots: x on (a, b), y on
(b, a). -/
abbrev Pair := Int  $\times$  Int

/-- The exchange of two identical particles. -/
def swap (p : Pair) : Pair := (p.2, p.1)

theorem swap_involutive (p : Pair) : swap (swap p) = p := rfl

/-- The exchange carrier: the antisymmetric part to the side, the
symmetric part to the height. -/
def exchangeSeat (p : Pair) : Pt := (1 + (p.1 - p.2), p.1 + p.2)

/-- THE EXCHANGE IS THE FOLD: the exchange carrier is equivariant. -/
theorem exchange_carrier_equivariant : Equivariant exchangeSeat swap
fold := by
  intro p
  obtain (x, y) := p
  show ((1 + (y - x), y + x) : Pt) = (2 - (1 + (x - y)), x + y)
  rw [pe]; exact (by omega, by omega)

/-- Bosons are the line: a pair amplitude lands on the line exactly
when it is symmetric. -/
theorem symmetric_on_the_line (p : Pair) : OnLine (exchangeSeat p)  $\leftrightarrow$ 
swap p = p := by
  obtain (x, y) := p
  show 1 + (x - y) = 1  $\leftrightarrow$  ((y, x) : Pair) = (x, y)
  rw [pe]; constructor
  · intro h; exact (by omega, by omega)
  · intro (h, _); omega

/-- The symmetric pairs, carried by the functor of the seat, land on
the line. -/
theorem exchange_carrier_lands :
   $\exists$  C : Carrier { p : Pair // swap p = p },  $\forall$  w, C.l w = exchangeSeat
  w.1 :=
  equivariant_carrier_lands exchangeSeat swap
  exchange_carrier_equivariant

/-- Fermions stand at height zero: an antisymmetric amplitude has no
symmetric part, and off the
line unless it vanishes. -/
theorem antisymmetric_at_height_zero (x : Int) (h : x  $\neq$  0) :
  (exchangeSeat (x, -x)).2 = 0  $\wedge$   $\neg$  OnLine (exchangeSeat (x, -x)) :=
  by
  refine (by show x + -x = 0; omega, ?_)
  show  $\neg$  (1 + (x - -x) = 1); omega

/-- The record keeps the symmetric part and forgets the antisymmetric
part: a pair and its exchange
are two states with one record. -/
theorem exchange_two_worlds (p : Pair) (h : p.1  $\neq$  p.2) :
  swap p  $\neq$  p  $\wedge$  reg (exchangeSeat (swap p)) = reg (exchangeSeat p) :=
  by
  refine {fun e  $\Rightarrow$  h (congrArg Prod.snd e), ?_}
  rw [exchange_carrier_equivariant]; rfl

/-! ## Part V. Pauli exclusion: the odd reading is blind on the fixed
set -/

/-- PAULI EXCLUSION: an antisymmetric amplitude vanishes wherever the
exchange fixes the
configuration, both particles in one state. -/
theorem pauli_exclusion { $\alpha$  : Type} ( $\psi$  :  $\alpha \rightarrow \alpha \rightarrow$  Int) (hanti :  $\forall$  a b,
 $\psi$  a b = - $\psi$  b a) (a :  $\alpha$ ) :
   $\psi$  a a = 0 := by
  have := hanti a a
  omega

/-- The Slater form f(a)g(b) - f(b)g(a) is antisymmetric and vanishes
on the diagonal. -/
theorem slater_vanishes_on_diagonal { $\alpha$  : Type} (f g :  $\alpha \rightarrow$  Int) (a :  $\alpha$ ) :
  f a * g a - f a * g a = 0 := by omega

/-- Bosons may share: a symmetric product f(a)f(b) is nonzero on the
diagonal wherever f is. -/
theorem bosons_may_share { $\alpha$  : Type} (f :  $\alpha \rightarrow$  Int) (a :  $\alpha$ ) (h : f a  $\neq$ 
0) : f a * f a  $\neq$  0 :=
  Int.mul_ne_zero h h

/-- THE FERMION'S OCCUPATION IS A BIT: an occupation number equal to
its own square is 0 or 1. -/
theorem fermion_occupation_is_a_bit (n : Int) : n * n = n  $\leftrightarrow$  n = 0  $\vee$  n =
1 := by
  constructor
  · intro h
    have h2 : n * (n - 1) = 0 := by simp only [Int.mul_sub,
    Int.mul_one]; omega
    rcases Int.mul_eq_zero.mp h2 with h3 | h3
    · left; exact h3
    · right; omega
  · intro h; rcases h with h | h <;> subst h <;> decide

/-! ## Part VI. The Return: a full turn is minus one on spinors -/

structure Q4 where
  r : Int
  i : Int
  j : Int
  k : Int
  deriving DecidableEq, Repr

def qmul (a b : Q4) : Q4 :=
  (a.r*b.r - a.i*b.i - a.j*b.j - a.k*b.k, a.r*b.i + a.i*b.r + a.j*b.k -
  a.k*b.j,
  a.r*b.j - a.i*b.k + a.j*b.r + a.k*b.i, a.r*b.k + a.i*b.j - a.j*b.i +
  a.k*b.r)
def qneg (a : Q4) : Q4 := (-a.r, -a.i, -a.j, -a.k)
def qconj (a : Q4) : Q4 := (a.r, -a.i, -a.j, -a.k)
def qone : Q4 := (1, 0, 0, 0)
def qi : Q4 := (0, 1, 0, 0)
def qj : Q4 := (0, 0, 1, 0)
def qk : Q4 := (0, 0, 0, 1)

theorem qext {a b : Q4} (h1 : a.r = b.r) (h2 : a.i = b.i) (h3 : a.j =
b.j) (h4 : a.k = b.k) :
  a = b := by
  cases a; cases b; simp_all

theorem qmul_neg_left (a b : Q4) : qmul (qneg a) b = qneg (qmul a b) :=
by
  apply qext <;> simp only [qmul, qneg, Int.neg_mul] <;> omega
theorem qmul_neg_right (a b : Q4) : qmul a (qneg b) = qneg (qmul a b)
:= by
  apply qext <;> simp only [qmul, qneg, Int.mul_neg] <;> omega
theorem qconj_neg (a : Q4) : qconj (qneg a) = qneg (qconj a) := rfl
theorem qneg_neg (a : Q4) : qneg (qneg a) = a := by
  apply qext <;> simp only [qneg, Int.neg_neg]
theorem qmul_one_left (a : Q4) : qmul qone a = a := by
  apply qext <;> simp only [qmul, qone] <;> omega

/-- The Return: i j k = -1, the quaternion form of a full turn. -/
theorem the_return : qmul (qmul qi qj) qk = qneg qone := by decide

/-- A FULL TURN IS MINUS ONE ON SPINORS: the full-turn element -1
negates every spinor. -/
theorem full_turn_negates_spinors ( $\psi$  : Q4) : qmul (qneg qone)  $\psi$  = qneg
 $\psi$  := by
  rw [qmul_neg_left, qmul_one_left]

/-- A full turn is the identity on vectors, which it acts on by
conjugation. -/
theorem full_turn_fixes_vectors (v : Q4) : qmul (qmul (qneg qone) v)
(qconj (qneg qone)) = v := by
  rw [qmul_neg_left, qconj_neg, qmul_neg_right, qmul_neg_left,
  qneg_neg, qmul_one_left]
  apply qext <;> simp only [qmul, qconj, qone] <;> omega

/-- Two full turns are the identity on spinors. -/
theorem two_turns_are_identity : qmul (qneg qone) (qneg qone) = qone :=
by decide

/-- THE DOUBLE COVER IS TWO TO ONE: q and -q rotate every vector alike.
The kernel of the cover
is {1, -1}, one bit. -/
theorem double_cover_two_to_one (q v : Q4) :
  qmul (qmul (qneg q) v) (qconj (qneg q)) = qmul (qmul q v) (qconj q)
:= by
  rw [qmul_neg_left, qconj_neg, qmul_neg_right, qmul_neg_left,
  qneg_neg]

/-- The exchange sign of spin n/2, read as the sign of a full turn:
(-1)n. -/
def turnSign : Nat  $\rightarrow$  Int
| 0  $\Rightarrow$  1
| n + 1  $\Rightarrow$  - turnSign n

theorem turnSign_val (n : Nat) : turnSign n = if n % 2 = 0 then 1 else
-1 := by
  induction n with
  | zero  $\Rightarrow$  rfl
  | succ n ih  $\Rightarrow$ 
    show - turnSign n = _
    rw [ih]
    by_cases h : n % 2 = 0

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· have h2 : (n + 1) % 2 ≠ 0 := by omega
simp [h, h2]
· have h2 : (n + 1) % 2 = 0 := by omega
simp [h, h2]

/-- Half-integer spin turns to minus one; integer spin to plus one.
With the spin-statistics
theorem carried, the first are fermions and the second bosons. -/
theorem half_integer_spin_is_odd (n : Nat) : turnSign n = -1 ↔ n % 2 =
1 := by
rw [turnSign_val]
by_cases h : n % 2 = 0
· simp [h]
· have h1 : n % 2 = 1 := by omega
simp [h, h1]

/-! ## Part VII. Statistics is one bit in three dimensions -/

/-- STATISTICS IS ONE BIT: an exchange sign whose square is the
identity, as a double exchange is
in three dimensions, is +1 or -1. -/
theorem statistics_one_bit (e : Int) (h : e * e = 1) : e = 1 ∨ e = -1
:= by
have h2 : (e - 1) * (e + 1) = 0 := by
simp only [Int.sub_mul, Int.mul_add, Int.mul_one, Int.one_mul];
omega
rcases Int.mul_eq_zero.mp h2 with h3 | h3
· left; omega
· right; omega

/-- A quarter phase i squares to -1, so it is excluded in three
dimensions and admitted only in the
plane, where a double exchange need not be the identity: the anyon.
-/
theorem anyone_needs_the_plane : qmul qi qi = qneg qone ∧ qmul qi qi ≠
qone := by decide

/-! ## Part VIII. The record: bunching and antibunching read the bit -/

/-- The coincidence amplitude of two identical particles at a balanced
splitter, doubled: t = 1,
r = i, so t·t + ε·r·r = 1 - ε. -/
def coincidence (e : Int) : Int := 1 - e

/-- BOSONS BUNCH, FERMIONS PART: the coincidence vanishes for bosons
and is full for fermions. -/
theorem bosons_bunch_fermions_part : coincidence 1 = 0 ∧ coincidence
(-1) = 2 := by decide

/-- Orbitals f = (1, 1) and g = (1, -1) on two sites, and the pair
amplitudes f(a)g(b) ± f(b)g(a). -/
def fo (a : Bool) : Int := 1
def go (a : Bool) : Int := if a then -1 else 1
def pairAmp (e : Int) (a b : Bool) : Int := fo a * go b + e * (fo b *
go a)
def oneBody (e : Int) (a : Bool) : Int := pairAmp e a false * pairAmp e
a false + pairAmp e a true * pairAmp e a true

/-- THE ONE-BODY RECORD DOES NOT READ THE STATISTICS: the boson pair
and the fermion pair have one
density at every site and differ at coincidence, where the pair
reading reads the bit. -/
theorem statistics_unread_by_one_body :
oneBody 1 false = oneBody (-1) false ∧ oneBody 1 true = oneBody
(-1) true ∧
pairAmp 1 false false * pairAmp 1 false false = 4 ∧
pairAmp (-1) false false * pairAmp (-1) false false = 0 := by
decide

/-! ## Part IX. The capstone -/

/-- SPIN AND STATISTICS ON THE SEAT. The exchange is the fold under the
exchange carrier; bosons
are the line and fermions stand at height zero off it; a pair and
its exchange share one
record; an antisymmetric amplitude vanishes on the exchange's fixed
configurations, so a
fermion's occupation is one bit; a full turn negates spinors and
fixes vectors, the double
cover being two to one; an exchange sign whose square is the
identity is ±1; bosons bunch and
fermions part; the one-body record does not read the statistics. -/
theorem spin_and_statistics_on_the_seat :
Equivariant exchangeSeat swap fold ∧
(∀ p : Pair, OnLine (exchangeSeat p) ↔ swap p = p) ∧
(∀ p : Pair, p.1 ≠ p.2 → reg (exchangeSeat (swap p)) = reg
(exchangeSeat p)) ∧
(∀ (ψ : Bool → Bool → Int), (∀ a b, ψ a b = -ψ b a) → ∀ a, ψ a a
= 0) ∧

```

'SPHYS.SpinStatistics.pe' does not depend on any axioms
'SPHYS.SpinStatistics.fold_involutive' depends on axioms: [propext,
Quot.sound]
'SPHYS.SpinStatistics.seat_fixed_line' depends on axioms: [propext,
Quot.sound]

```

(∀ n : Int, n * n = n ↔ n = 0 ∨ n = 1) ∧
(∀ ψ : Q4, qmul (qneg qone) ψ = qneg ψ) ∧
(∀ q v : Q4, qmul (qmul (qneg q) v) (qconj (qneg q)) = qmul (qmul q
v) (qconj q)) ∧
(∀ e : Int, e * e = 1 → e = 1 ∨ e = -1) ∧
(coincidence 1 = 0 ∧ coincidence (-1) = 2) ∧
(oneBody 1 false = oneBody (-1) false ∧ oneBody 1 true = oneBody
(-1) true) :=
(exchange_carrier_equivariant, symmetric_on_the_line, fun p h =>
(exchange_two_worlds p h).2,
fun ψ h a => pauli_exclusion ψ h a, fermion_occupation_is_a_bit,
full_turn_negates_spinors,
double_cover_two_to_one, statistics_one_bit,
bosons_bunch_fermions_part,
(statistics_unread_by_one_body.1,
statistics_unread_by_one_body.2.1))

```

end SPHYS.SpinStatistics

```

#print axioms SPHYS.SpinStatistics.pe
#print axioms SPHYS.SpinStatistics.fold_involutive
#print axioms SPHYS.SpinStatistics.seat_fixed_line
#print axioms SPHYS.SpinStatistics.reg_lands
#print axioms SPHYS.SpinStatistics.reg_fixes_iff
#print axioms SPHYS.SpinStatistics.reg_forgets_side
#print axioms SPHYS.SpinStatistics.least_erase_uff_value
#print axioms SPHYS.SpinStatistics.equivariant_id
#print axioms SPHYS.SpinStatistics.equivariant_comp
#print axioms SPHYS.SpinStatistics.fix_functorial
#print axioms SPHYS.SpinStatistics.fold_global_seat
#print axioms SPHYS.SpinStatistics.equivariant_carrier_lands
#print axioms SPHYS.SpinStatistics.value_on_image
#print axioms SPHYS.SpinStatistics.kinetic_crossing
#print axioms SPHYS.SpinStatistics.off_locus_pair
#print axioms SPHYS.SpinStatistics.ee
#print axioms SPHYS.SpinStatistics.mirror_involutive
#print axioms SPHYS.SpinStatistics.psi_phi
#print axioms SPHYS.SpinStatistics.phi_psi
#print axioms SPHYS.SpinStatistics.phi_injective
#print axioms SPHYS.SpinStatistics.phi_equivariant
#print axioms SPHYS.SpinStatistics.mirror_fixed_iff_real
#print axioms SPHYS.SpinStatistics.phi_line_iff_real
#print axioms SPHYS.SpinStatistics.stationary_iff_real
#print axioms SPHYS.SpinStatistics.line_is_stationary
#print axioms SPHYS.SpinStatistics.measured_on_line
#print axioms SPHYS.SpinStatistics.no_measurement_off_line
#print axioms SPHYS.SpinStatistics.energy_carrier_lands
#print axioms SPHYS.SpinStatistics.bar_involutive
#print axioms SPHYS.SpinStatistics.bar_fixed_iff_neutral
#print axioms SPHYS.SpinStatistics.odd_charges_even_energy
#print axioms SPHYS.SpinStatistics.gravity_reads_no_charge_bit
#print axioms SPHYS.SpinStatistics.involutions_commute
#print axioms SPHYS.SpinStatistics.tmirror_involutive
#print axioms SPHYS.SpinStatistics.joint_fixed_iff
#print axioms SPHYS.SpinStatistics.charge_carrier_equivariant
#print axioms SPHYS.SpinStatistics.pair_lands_once
#print axioms SPHYS.SpinStatistics.tmirror_equivariant
#print axioms SPHYS.SpinStatistics.neutral_charge_on_line
#print axioms SPHYS.SpinStatistics.pair_on_the_edges
#print axioms SPHYS.SpinStatistics.stable_lands
#print axioms SPHYS.SpinStatistics.swap_involutive
#print axioms SPHYS.SpinStatistics.exchange_carrier_equivariant
#print axioms SPHYS.SpinStatistics.symmetric_on_the_line
#print axioms SPHYS.SpinStatistics.exchange_carrier_lands
#print axioms SPHYS.SpinStatistics.antisymmetric_at_height_zero
#print axioms SPHYS.SpinStatistics.exchange_two_worlds
#print axioms SPHYS.SpinStatistics.pauli_exclusion
#print axioms SPHYS.SpinStatistics.slater_vanishes_on_diagonal
#print axioms SPHYS.SpinStatistics.bosons_may_share
#print axioms SPHYS.SpinStatistics.fermion_occupation_is_a_bit
#print axioms SPHYS.SpinStatistics.qext
#print axioms SPHYS.SpinStatistics.qmul_neg_left
#print axioms SPHYS.SpinStatistics.qmul_neg_right
#print axioms SPHYS.SpinStatistics.qconj_neg
#print axioms SPHYS.SpinStatistics.qneg_neg
#print axioms SPHYS.SpinStatistics.qmul_one_left
#print axioms SPHYS.SpinStatistics.the_return
#print axioms SPHYS.SpinStatistics.full_turn_negates_spinors
#print axioms SPHYS.SpinStatistics.full_turn_fixes_vectors
#print axioms SPHYS.SpinStatistics.two_turns_are_identity
#print axioms SPHYS.SpinStatistics.double_cover_two_to_one
#print axioms SPHYS.SpinStatistics.turnSign_val
#print axioms SPHYS.SpinStatistics.half_integer_spin_is_odd
#print axioms SPHYS.SpinStatistics.statistics_one_bit
#print axioms SPHYS.SpinStatistics.anyone_needs_the_plane
#print axioms SPHYS.SpinStatistics.bosons_bunch_fermions_part
#print axioms SPHYS.SpinStatistics.statistics_unread_by_one_body
#print axioms SPHYS.SpinStatistics.spin_and_statistics_on_the_seat

```

'SPHYS.SpinStatistics.reg_lands' does not depend on any axioms
'SPHYS.SpinStatistics.reg_fixes_iff' depends on axioms: [propext]
'SPHYS.SpinStatistics.reg_forgets_side' does not depend on any axioms
'SPHYS.SpinStatistics.least_erase_uff_value' depends on axioms:
[propext]

```

'SPHYS.SpinStatistics.equivariant_id' does not depend on any axioms
'SPHYS.SpinStatistics.equivariant_comp' does not depend on any axioms
'SPHYS.SpinStatistics.fix_functorial' does not depend on any axioms
'SPHYS.SpinStatistics.fold_global_seat' depends on axioms: [propext,
Quot.sound]
'SPHYS.SpinStatistics.equivariant_carrier_lands' depends on axioms:
[propext, Quot.sound]
'SPHYS.SpinStatistics.value_on_image' does not depend on any axioms
'SPHYS.SpinStatistics.kinetic_crossing' does not depend on any axioms
'SPHYS.SpinStatistics.off_locus_pair' depends on axioms: [propext,
Quot.sound]
'SPHYS.SpinStatistics.ee' does not depend on any axioms
'SPHYS.SpinStatistics.mirror_involutive' depends on axioms: [propext,
Quot.sound]
'SPHYS.SpinStatistics.psi_phi' depends on axioms: [propext, Quot.sound]
'SPHYS.SpinStatistics.phi_psi' depends on axioms: [propext, Quot.sound]
'SPHYS.SpinStatistics.phi_injective' depends on axioms: [propext,
Quot.sound]
'SPHYS.SpinStatistics.phi_equivariant' depends on axioms: [propext,
Quot.sound]
'SPHYS.SpinStatistics.mirror_fixed_iff_real' depends on axioms:
[propext, Quot.sound]
'SPHYS.SpinStatistics.phi_line_iff_real' depends on axioms: [propext,
Quot.sound]
'SPHYS.SpinStatistics.stationary_iff_real' depends on axioms: [propext]
'SPHYS.SpinStatistics.line_is_stationary' depends on axioms: [propext,
Quot.sound]
'SPHYS.SpinStatistics.measured_on_line' does not depend on any axioms
'SPHYS.SpinStatistics.no_measurement_off_line' does not depend on any
axioms
'SPHYS.SpinStatistics.energy_carrier_lands' depends on axioms:
[propext, Quot.sound]
'SPHYS.SpinStatistics.bar_involutive' does not depend on any axioms
'SPHYS.SpinStatistics.bar_fixed_iff_neutral' depends on axioms:
[propext, Quot.sound]
'SPHYS.SpinStatistics.odd_charges_even_energy' does not depend on any
axioms
'SPHYS.SpinStatistics.gravity_reads_no_charge_bit' does not depend on
any axioms
'SPHYS.SpinStatistics.involutions_commute' does not depend on any
axioms
'SPHYS.SpinStatistics.tmirror_involutive' depends on axioms: [propext,
Quot.sound]
'SPHYS.SpinStatistics.joint_fixed_iff' depends on axioms: [propext,
Quot.sound]
'SPHYS.SpinStatistics.charge_carrier_equivariant' depends on axioms:
[propext, Quot.sound]
'SPHYS.SpinStatistics.pair_lands_once' does not depend on any axioms
'SPHYS.SpinStatistics.tmirror_equivariant' depends on axioms: [propext,
Quot.sound]
'SPHYS.SpinStatistics.neutral_charge_on_line' depends on axioms:
[propext, Quot.sound]

```

```

'SPHYS.SpinStatistics.pair_on_the_edges' does not depend on any axioms
'SPHYS.SpinStatistics.stable_lands' depends on axioms: [propext,
Quot.sound]
'SPHYS.SpinStatistics.swap_involutive' does not depend on any axioms
'SPHYS.SpinStatistics.exchange_carrier_equivariant' depends on axioms:
[propext, Quot.sound]
'SPHYS.SpinStatistics.symmetric_on_the_line' depends on axioms:
[propext, Quot.sound]
'SPHYS.SpinStatistics.exchange_carrier_lands' depends on axioms:
[propext, Quot.sound]
'SPHYS.SpinStatistics.antisymmetric_at_height_zero' depends on axioms:
[propext, Quot.sound]
'SPHYS.SpinStatistics.exchange_two_worlds' depends on axioms: [propext,
Quot.sound]
'SPHYS.SpinStatistics.pauli_exclusion' depends on axioms: [propext,
Quot.sound]
'SPHYS.SpinStatistics.slater_vanishes_on_diagonal' depends on axioms:
[propext, Quot.sound]
'SPHYS.SpinStatistics.bosons_may_share' depends on axioms: [propext]
'SPHYS.SpinStatistics.fermion_occupation_is_a_bit' depends on axioms:
[propext, Quot.sound]
'SPHYS.SpinStatistics.qext' depends on axioms: [propext]
'SPHYS.SpinStatistics.qmul_neg_left' depends on axioms: [propext,
Quot.sound]
'SPHYS.SpinStatistics.qmul_neg_right' depends on axioms: [propext,
Quot.sound]
'SPHYS.SpinStatistics.qconj_neg' does not depend on any axioms
'SPHYS.SpinStatistics.qneg_neg' depends on axioms: [propext]
'SPHYS.SpinStatistics.qmul_one_left' depends on axioms: [propext,
Quot.sound]
'SPHYS.SpinStatistics.the_return' does not depend on any axioms
'SPHYS.SpinStatistics.full_turn_negates_spinors' depends on axioms:
[propext, Quot.sound]
'SPHYS.SpinStatistics.full_turn_fixes_vectors' depends on axioms:
[propext, Quot.sound]
'SPHYS.SpinStatistics.two_turns_are_identity' does not depend on any
axioms
'SPHYS.SpinStatistics.double_cover_two_to_one' depends on axioms:
[propext, Quot.sound]
'SPHYS.SpinStatistics.turnSign_val' depends on axioms: [propext,
Quot.sound]
'SPHYS.SpinStatistics.half_integer_spin_is_odd' depends on axioms:
[propext, Quot.sound]
'SPHYS.SpinStatistics.statistics_one_bit' depends on axioms: [propext,
Quot.sound]
'SPHYS.SpinStatistics.anyon_needs_the_plane' does not depend on any
axioms
'SPHYS.SpinStatistics.bosons_bunch_fermions_part' does not depend on
any axioms
'SPHYS.SpinStatistics.statistics_unread_by_one_body' does not depend on
any axioms
'SPHYS.SpinStatistics.spin_and_statistics_on_the_seat' depends on
axioms: [propext, Quot.sound]

```

APPENDIX B: SPINSTAT_TWIN.F90 AND ITS RUN

```

! SpinStat_Twin.f90 · the executed twin of SPHYS_Spin_Statistics.lean ·
Fortran 2018.
! Build (sealed): gfortran -std=f2018 -O2 -fno-fast-math
-ffp-contract=off SpinStat_Twin.f90
! Blocks: A the exchange carrier; B Pauli exclusion; C the Return and
the double cover; D one bit in
! three dimensions; E bunching and antibunching; F the periodic table
from the bit; G degeneracy;
! Chandrasekhar's mass; H condensation: London's temperature; I spin
and statistics on the table.
program spinstat_twin
implicit none
integer, parameter :: dp = selected_real_kind(15, 307)
integer, parameter :: i8 = selected_int_kind(18)
real(dp), parameter :: PI = 3.141592653589793_dp, HBAR =
1.054571817e-34_dp, CL = 299792458.0_dp
real(dp), parameter :: G = 6.67430e-11_dp, MH = 1.6735575e-27_dp,
MSUN = 1.98847e30_dp
real(dp), parameter :: KB = 1.380649e-23_dp, AMU =
1.66053906660e-27_dp
integer :: checks, fails
integer(i8) :: seed
checks = 0; fails = 0; seed = 20260930_i8
call block_a(); call block_b(); call block_c(); call block_d(); call
block_e()
call block_f(); call block_g(); call block_h(); call block_i()
write(*, '(a,i0,a,i0,a)') ' BATTERY-JSON: {"checks":', checks,
', "failures":', fails, '}'
if (fails > 0) error stop 1
contains
subroutine check(name, ok)
character(*), intent(in) :: name
logical, intent(in) :: ok
checks = checks + 1
if (.not. ok) fails = fails + 1
write(*, '(a,a)') merge(' PASS ', ' FAIL ', ok), name

```

```

end subroutine check

function rnd() result(r)
real(dp) :: r
seed = modulo(16807_i8*seed, 2147483647_i8)
r = real(seed, dp)/2147483647.0_dp
end function rnd

subroutine block_a()
integer :: x, y, n, neq, nline, nzero, npair
write(*, '(a)') 'A · the exchange is the fold: bosons on the line,
fermions at height zero'
n = 0; neq = 0; nline = 0; nzero = 0; npair = 0
do x = -30, 30
do y = -30, 30
n = n + 1
! seat of (x, y) is (1 + (x - y), x + y); seat of the swap is
(1 + (y - x), y + x)
if (1 + (y - x) == 2 - (1 + (x - y)) .and. y + x == x + y) neq =
neq + 1
if ((1 + (x - y) == 1) .eqv. (x == y)) nline = nline + 1
if (x == -y) then
if (x + y == 0 .and. ((1 + (x - y) /= 1) .eqv. (x /= 0)))
nzero = nzero + 1
end if
if (x /= y .and. x + y == y + x) npair = npair + 1
end do
end do
call check('equivariance on 3721 pair amplitudes: the exchange is
the fold', neq == n)
call check('a pair amplitude is on the line exactly when it is
symmetric (3721)', nline == n)
call check('antisymmetric amplitudes stand at height zero, off the
line unless zero (61)', nzero == 61)
call check('a pair and its exchange share one record (3660
unsymmetric pairs)', npair == n - 61)

```

```

end subroutine block_a

subroutine block_b()
integer :: t, a, b, nd, nsl, nbit, nn
real(dp) :: psi(6,6), f(6), gg(6), m
write(*,'(a)') 'B · Pauli exclusion: the odd amplitude is blind on
the diagonal; the occupation is a bit'
nd = 0; nsl = 0
do t = 1, 1000
do a = 1, 6
f(a) = rnd() - 0.5_dp; gg(a) = rnd() - 0.5_dp
do b = 1, 6
psi(a,b) = rnd() - 0.5_dp
end do
end do
psi = psi - transpose(psi)
if (all([(psi(a,a) == 0.0_dp, a = 1, 6)]) nd = nd + 1
m = 0.0_dp
do a = 1, 6
m = max(m, abs(f(a)*gg(a) - f(a)*gg(a)))
end do
if (m == 0.0_dp) nsl = nsl + 1
end do
call check('1000 random antisymmetric amplitudes vanish exactly on
every diagonal entry', nd == 1000)
call check('1000 random Slater amplitudes f(a)g(b) - f(b)g(a)
vanish at a = b', nsl == 1000)
nbit = 0
do nn = -50, 50
if ((nn*nn == nn) .eqv. (nn == 0 .or. nn == 1)) nbit = nbit + 1
end do
call check('an occupation equal to its own square is 0 or 1 (101 of
101)', nbit == 101)
end subroutine block_b

pure function qm(a, b) result(c)
real(dp), intent(in) :: a(4), b(4)
real(dp) :: c(4)
c(1) = a(1)*b(1) - a(2)*b(2) - a(3)*b(3) - a(4)*b(4)
c(2) = a(1)*b(2) + a(2)*b(1) + a(3)*b(4) - a(4)*b(3)
c(3) = a(1)*b(3) - a(2)*b(4) + a(3)*b(1) + a(4)*b(2)
c(4) = a(1)*b(4) + a(2)*b(3) - a(3)*b(2) + a(4)*b(1)
end function qm

pure function qc(a) result(c)
real(dp), intent(in) :: a(4)
real(dp) :: c(4)
c = [a(1), -a(2), -a(3), -a(4)]
end function qc

subroutine block_c()
integer :: t
real(dp) :: u(3), q2(4), q4(4), v(4), w(4), psi(4), s(4), e2, e4,
ev, es, ec, q(4), r1(4), r2(4)
write(*,'(a)') 'C · the Return: a full turn is -1 on spinors, +1 on
vectors; the cover is two to one'
e2 = 0.0_dp; e4 = 0.0_dp; ev = 0.0_dp; es = 0.0_dp; ec = 0.0_dp
do t = 1, 1000
u = [rnd() - 0.5_dp, rnd() - 0.5_dp, rnd() - 0.5_dp]; u =
u/sqrt(sum(u*u))
q2 = [cos(PI), sin(PI)*u(1), sin(PI)*u(2), sin(PI)*u(3)]
q4 = [cos(2.0_dp*PI), sin(2.0_dp*PI)*u(1), sin(2.0_dp*PI)*u(2),
sin(2.0_dp*PI)*u(3)]
e2 = max(e2, maxval(abs(q2 - [-1.0_dp, 0.0_dp, 0.0_dp, 0.0_dp])))
e4 = max(e4, maxval(abs(q4 - [1.0_dp, 0.0_dp, 0.0_dp, 0.0_dp])))
v = [0.0_dp, rnd() - 0.5_dp, rnd() - 0.5_dp, rnd() - 0.5_dp]
w = qm(q2, v), qc(q2); ev = max(ev, maxval(abs(w - v)))
psi = [rnd() - 0.5_dp, rnd() - 0.5_dp, rnd() - 0.5_dp, rnd() -
0.5_dp]
s = qm(q2, psi); es = max(es, maxval(abs(s + psi)))
q = [rnd() - 0.5_dp, rnd() - 0.5_dp, rnd() - 0.5_dp, rnd() -
0.5_dp]; q = q/sqrt(sum(q*q))
r1 = qm(qm(q, v), qc(q)); r2 = qm(qm(-q, v), qc(-q)); ec =
max(ec, maxval(abs(r1 - r2)))
end do
write(*,'(a,es9.2,a,es9.2,a,es9.2)') ' full turn: |q - (-1)| <= ',
e2, ' ; vectors return to ', ev, &
' ; spinors to -psi within ', es
call check('a full turn about 1000 random axes is -1 on spinors and
the identity on vectors', &
e2 < 1.0e-15_dp .and. ev < 1.0e-15_dp .and. es <
1.0e-15_dp)
call check('two full turns are the identity, and q and -q rotate
every vector alike (1000)', &
e4 < 1.0e-15_dp .and. ec < 1.0e-15_dp)
end subroutine block_c

subroutine block_d()
integer :: t, n
real(dp) :: th, re, im
write(*,'(a)') 'D · statistics is one bit in three dimensions; the
anyon needs the plane'
n = 0
do t = 0, 3599
th = 2.0_dp*PI*real(t, dp)/3600.0_dp
re = cos(2.0_dp*th); im = sin(2.0_dp*th)
if (abs(re - 1.0_dp) < 1.0e-12_dp .and. abs(im) < 1.0e-12_dp) n =
n + 1
end do
write(*,'(a,i0,a)') ' exchange phases on a grid of 3600 whose
square is the identity: ', n, ' (0 and pi)'
call check('only the phases +1 and -1 square to the identity', n ==
2)
call check('the fractional phase pi/3 of the nu = 1/3 anyon does
not square to the identity', &
abs(cos(2.0_dp*PI/3.0_dp) - 1.0_dp) > 0.5_dp)
end subroutine block_d

subroutine block_e()
integer :: t
real(dp) :: x, pb, pf, worst, gb, gf
write(*,'(a)') 'E · the record reads the bit: bosons bunch,
fermions part'
worst = 0.0_dp
do t = 0, 100
x = real(t, dp)/100.0_dp
pb = 0.5_dp*(1.0_dp - x); pf = 0.5_dp*(1.0_dp + x)
worst = max(worst, abs(pb + pf - 1.0_dp))
end do
call check('at a balanced splitter, full overlap: bosons never
coincide, fermions always do', &
0.5_dp*(1.0_dp - 1.0_dp) == 0.0_dp .and. 0.5_dp*(1.0_dp
+ 1.0_dp) == 1.0_dp .and. worst < 1.0e-15_dp)
call thermal_g2(0.3_dp, gb, gf)
write(*,'(a,f9.6,a,f9.6)') ' one thermal mode at mean occupation
0.3: Bose g2 = ', gb, ' ; Fermi g2 = ', gf
call check('a thermal Bose mode bunches, g2 = 2 (helium-4), and a
Fermi mode antibunches, g2 = 0 (helium-3)', &
abs(gb - 2.0_dp) < 1.0e-12_dp .and. gf == 0.0_dp)
end subroutine block_e

subroutine thermal_g2(nbar, gb, gf)
real(dp), intent(in) :: nbar
real(dp), intent(out) :: gb, gf
integer :: n
real(dp) :: p, s1, s2, r
! Bose: geometric occupation with ratio r = nbar/(1 + nbar); Fermi:
occupation 0 or 1 only
r = nbar/(1.0_dp + nbar); s1 = 0.0_dp; s2 = 0.0_dp
do n = 0, 400
p = (1.0_dp - r)*r**n
s1 = s1 + real(n, dp)*p; s2 = s2 + real(n, dp)*real(n - 1, dp)*p
end do
gb = s2/(s1*s1)
s1 = 0.0_dp*(1.0_dp - nbar) + 1.0_dp*nbar
s2 = 0.0_dp*(1.0_dp - nbar) + 0.0_dp*nbar
gf = s2/(s1*s1)
end subroutine thermal_g2

subroutine block_f()
integer, parameter :: order(19,2) =
reshape([1,2,2,3,3,4,3,4,4,5,4,5,6,4,5,6,7,5,6,7, &
0,0,1,0,1,0,2,1,0,2,1]
,0,3,2,1,0,3,2,1],
[19,2])
integer :: k, total, nclos, closures(7)
write(*,'(a)') 'F · the periodic table from the bit: two states per
orbital, one per state'
total = 0; nclos = 0
do k = 1, 19
total = total + 2*(2*order(k,2) + 1)
if ((order(k,2) == 1 .or. (order(k,1) == 1 .and. order(k,2) ==
0)) .and. nclos < 7) then
nclos = nclos + 1; closures(nclos) = total
end if
end do
write(*,'(a,i7,i5)') ' shell closures from the Madelung order: ',
closures
call check('the noble gases stand at 2, 10, 18, 36, 54, 86, 118', &
all(closures == [2, 10, 18, 36, 54, 86, 118]))
end subroutine block_f

subroutine block_g()
real(dp) :: mch, omega3, mue
write(*,'(a)') 'G · degeneracy: the fermion bit holds a white dwarf
up to Chandrasekhar's mass'
omega3 = 2.01824_dp; mue = 2.0_dp
mch =
omega3*sqrt(3.0_dp*PI)/2.0_dp*(HBAR*CL/G)**1.5_dp/(mue*MH)**2/MSUN
write(*,'(a,f6.3,a)') ' Chandrasekhar mass for mu_e = 2: ', mch, '
solar masses'
call check('the degenerate-electron limit is 1.43 solar masses
(ideal, mu_e = 2)', abs(mch - 1.43_dp) < 0.02_dp)
end subroutine block_g

subroutine block_h()
real(dp) :: m4, n, tc
write(*,'(a)') 'H · condensation: the boson bit, London's estimate
for helium-4'
m4 = 4.002602_dp*AMU; n = 145.0_dp/m4
tc = 2.0_dp*PI*HBAR**2/(m4*KB)*(n/2.6123753_dp)**(2.0_dp/3.0_dp)
write(*,'(a,f6.3,a)') ' ideal Bose gas at liquid-helium density:
T_c = ', tc, ' K (the lambda point is 2.17 K)'

```

```

    call check('London''s ideal-gas temperature is 3.13 K, the scale of
    the lambda transition', abs(tc - 3.13_dp) < 0.05_dp)
end subroutine block_h

subroutine block_i()
  integer, parameter :: ntab = 12
  integer :: spin2(ntab), ferm(ntab), k, nok
  ! twice the spin, and the number of elementary fermion
  constituents, for
  ! e, p, n, photon, W, gluon, Higgs, pion, deuteron, helium-4 atom,
  helium-3 atom, Delta
  ! (quarks and electrons counted: the deuteron six quarks, helium-4
  twelve and two, helium-3 nine and two)

  spin2 = [1, 1, 1, 2, 2, 2, 0, 0, 2, 0, 1, 3]
  ferm = [1, 3, 3, 0, 0, 0, 0, 2, 6, 14, 11, 3]
  write(*, '(a)') 'I · spin and statistics on the table: the turn sign
  is the exchange sign'
  nok = 0
  do k = 1, ntab
    if (mod(spin2(k), 2) == mod(ferm(k), 2)) nok = nok + 1
  end do
  call check('twelve particles and atoms: half-integer spin exactly
  when an odd number of fermions', nok == ntab)
end subroutine block_i
end program spinstat_twin

A · the exchange is the fold: bosons on the line, fermions at height
zero
PASS  equivariance on 3721 pair amplitudes: the exchange is the fold
PASS  a pair amplitude is on the line exactly when it is symmetric
(3721)
PASS  antisymmetric amplitudes stand at height zero, off the line
unless zero (61)
PASS  a pair and its exchange share one record (3660 unsymmetric
pairs)
B · Pauli exclusion: the odd amplitude is blind on the diagonal; the
occupation is a bit
PASS  1000 random antisymmetric amplitudes vanish exactly on every
diagonal entry
PASS  1000 random Slater amplitudes f(a)g(b) - f(b)g(a) vanish at a =
b
PASS  an occupation equal to its own square is 0 or 1 (101 of 101)
C · the Return: a full turn is -1 on spinors, +1 on vectors; the cover
is two to one
full turn: |q - (-1)| <= 1.22E-16; vectors return to 2.22E-16;
spinors to -psi within 1.11E-16
PASS  a full turn about 1000 random axes is -1 on spinors and the
identity on vectors
PASS  two full turns are the identity, and q and -q rotate every
vector alike (1000)
D · statistics is one bit in three dimensions; the anyon needs the
plane
exchange phases on a grid of 3600 whose square is the identity: 2 (0
and pi)
PASS  only the phases +1 and -1 square to the identity

PASS  the fractional phase pi/3 of the nu = 1/3 anyon does not square
to the identity
E · the record reads the bit: bosons bunch, fermions part
PASS  a balanced splitter, full overlap: bosons never coincide,
fermions always do
one thermal mode at mean occupation 0.3: Bose g2 = 2.000000; Fermi
g2 = 0.000000
PASS  a thermal Bose mode bunches, g2 = 2 (helium-4), and a Fermi
mode antibunches, g2 = 0 (helium-3)
F · the periodic table from the bit: two states per orbital, one per
state
shell closures from the Madelung order:      2   10   18   36   54
86  118
PASS  the noble gases stand at 2, 10, 18, 36, 54, 86, 118
G · degeneracy: the fermion bit holds a white dwarf up to
Chandrasekhar's mass
Chandrasekhar mass for mu_e = 2: 1.434 solar masses
PASS  the degenerate-electron limit is 1.43 solar masses (ideal, mu_e
= 2)
H · condensation: the boson bit, London's estimate for helium-4
ideal Bose gas at liquid-helium density: T_c = 3.134 K (the lambda
point is 2.17 K)
PASS  London's ideal-gas temperature is 3.13 K, the scale of the
lambda transition
I · spin and statistics on the table: the turn sign is the exchange
sign
PASS  twelve particles and atoms: half-integer spin exactly when an
odd number of fermions
BATTERY-JSON: {"checks":17,"failures":0}

```

APPENDIX C: THE SEAT AND THE JUDGMENT

```

PhysOST 1.0.5p · RECEIPT · mode quick
TOOLCHAIN Lean (version 4.19.0, x86_64-unknown-linux-gnu, commit 6caae842e94, Release) · GNU Fortran (Ubuntu 13.3.0-6ubuntu2-24.04.1) 13.3.0
MANIFEST sha256 a4108cd26bcbe6ef, the files that ran
GROUND toolchain: Lean 4.19.0 and gfortran present · manifest: 31 files signed, all verified · proofs: 2 PhysOS Proofs read from II.10, record
lines complete · source screen: 15 kernels, 23 named premises, clean · citations: 253 laws named in Parts I and II, 253 resolved · quarantine: 6
retirements string-scanned, 3 held by review, clean
EXITS ground 0 · registerA RA_TOE_Thesis 0 · registerB Bridge_Final 0 · registerB Universal_Closure 0 · registerB One_Cut 0 · registerB
One_Cut_Resolution 0 · registerB One_Cut_Terminal 0 · registerB Only_The_Arrow_Remains 0 · registerB Physical_Closure 0 · registerB Forced_Closure
0 · registerB Closure_Executed 0 · registerB Armed_Seat 0 · registerB Census 0 · registerB Heat_Bridge 0 · registerB The_Loop 0 · registerB
RH_Least_Erasure 0 · witness One_Cut_Twin 0 · witness Physical_Closure_Twin 0 · witness Census_Twin 0 · witness Loop_Twin 0 · witness RH_Seal_Twin
0
REGISTER A {"checks":1123,"failures":0,"mode":"sealed"}
WITNESS 5 of 5 twins ran, each held to one battery line with zero failures
CHAIN · D0 83f872725ad9 -> D1 90fc702bbd1 -> D2 492e519daab9 -> D3 6618bec5164b
ELAPSED ground 1 s · Register A 3 s · Register B 34 s · witness 6 s · total 44 s (not hashed)
SEAT EARNED · this run

SCREEN forbidden: none | imports: 0 | axioms declared: 0
laws: 69
JUDGMENT refused 69 of 69 as proof failures | survived none | parse failures none | planted vacuous law survives its negation: True

```